Error bounds for non-smooth right-hand sides

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Problem statement

Find error bounds for the numerical integration of

$$y_s'(t) = f(t, y_s(t), s(t, y_s(t))), \quad (t \in [t_0, t_e]), \quad y_s(t_0) = y_0,$$

where s is "reasonably non-smooth".

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Find error bounds for the numerical integration of

$$y_s'(t) = f(t, y_s(t), s(t, y_s(t))), \quad (t \in [t_0, t_e]), \quad y_s(t_0) = y_0,$$

where s is "reasonably non-smooth".

Definition

A function s is reasonably non-smooth, if

1. $s \in C^k, k \in \mathbb{N}_0$,
2. $s^{(k+1)}$ discontinuous only at finitely many points and
3. s has uniformly bounded (weak) derivatives up to order $k+1$, i.e. given $L > 0$, $\|s^{(r)}\|_\infty \leq L$, $r = 0, 1, \ldots, k+1$.

Application

- Real-world data for u in the smooth system

$$y_u'(t) = f(t, y_u(t), u(t, y_u(t))), \quad (t \in [t_0, t_e]), \quad y_u(t_0) = y_0$$

  known only at finitely many times
- Real-world data available for y, but (potentially non-smooth) dynamics unknown (NeuralODE)
- Physical model is non-smooth (impact, switches, different time scales)



Towards error estimates, Arnold 2010

Theorem

The numerical integration of

$$y_s'(t) = f(t, y_s(t), s(t)), \quad (t \in [t_0, t_e]), \quad y_s(t_0) = y_0,$$

where $s \in C^k, (k = 0, 2)$, *is a linear or cubic spline of* $u(t)$, *yields local error estimates*

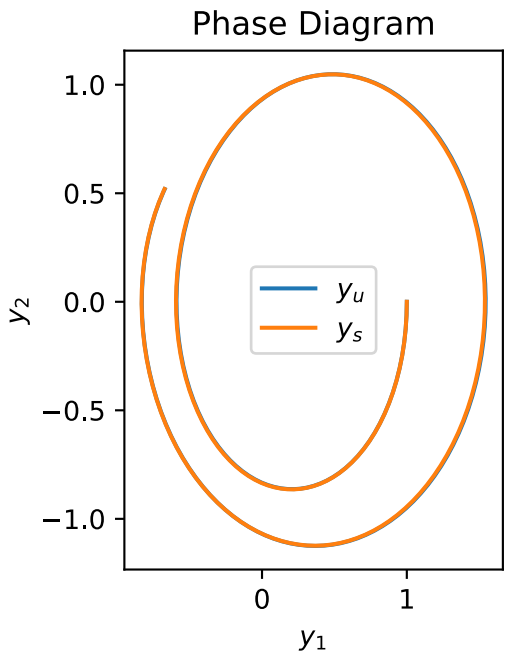
$$\|le_n\| = \mathcal{O}(h^{p+1}) \text{ and } \|le_n\| = \mathcal{O}(h^{p+1}) + \mathcal{O}(h^{k+2}).$$

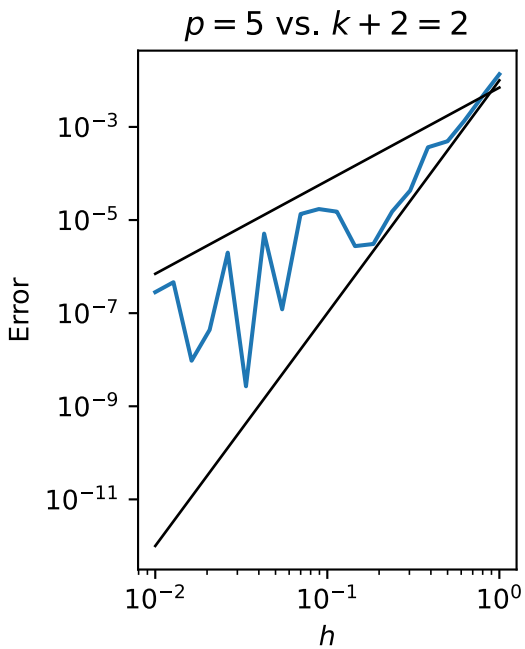
and since $s^{(k+1)}$ *discontinuous only at finitely many points the global error estimate*

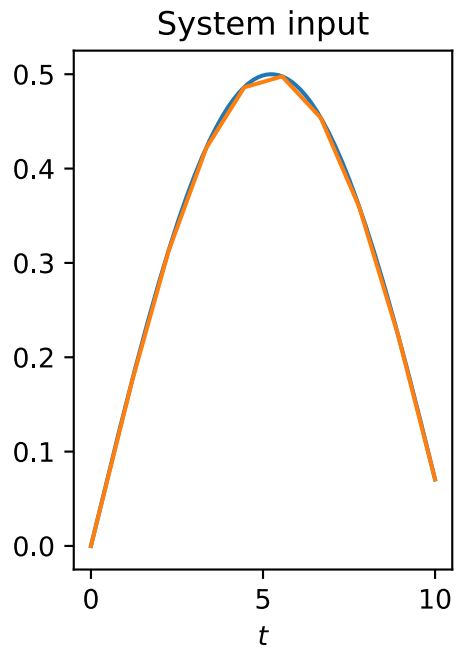
$$\|e_n\| = \mathcal{O}(h^p) + \mathcal{O}(h^{k+2}).$$

Excited Oscillator, 5th order RK, linear spline

$$y_1''(t) + y_1(t) = u(t),\ u(t) = \alpha \sin(\omega t), y_2 = y_1'.$$



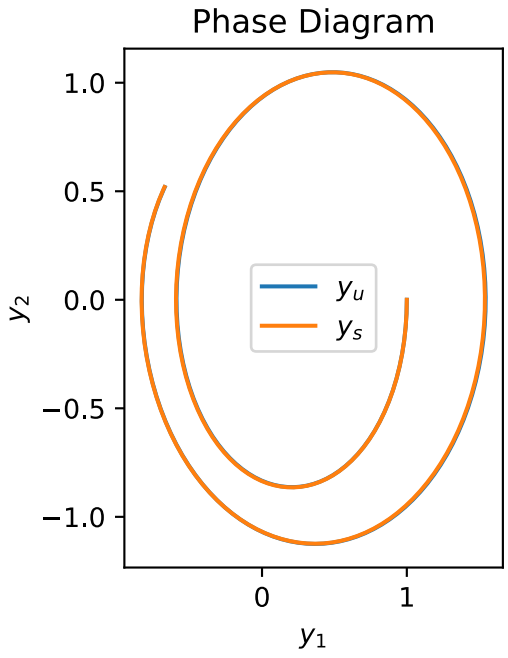


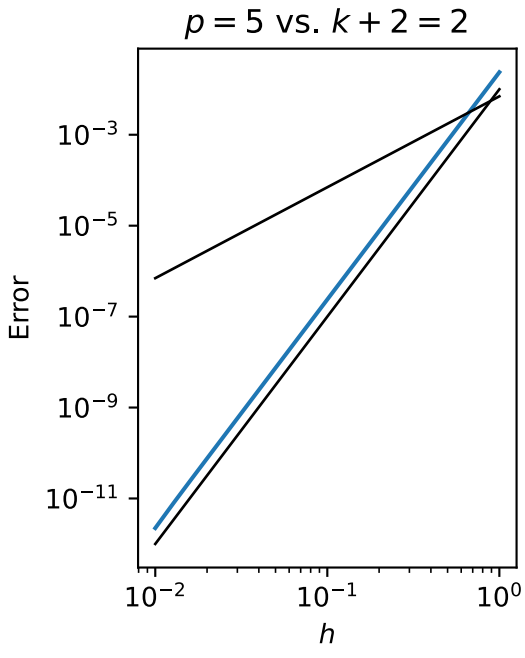


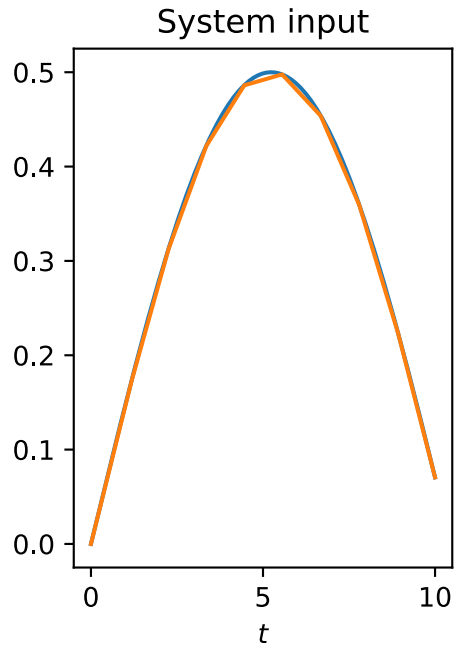
$$y_1(0) = 1,\ y_1'(0) = 0,\ y_1' = y_2, \alpha = 0.5, \omega = 0.3, t_0 = 0, t_e = 10.$$

Excited Oscillator, 5th order RK, step width control

$$y_1''(t) + y_1(t) = u(t),\ u(t) = \alpha \sin(\omega t), y_2 = y_1'.$$



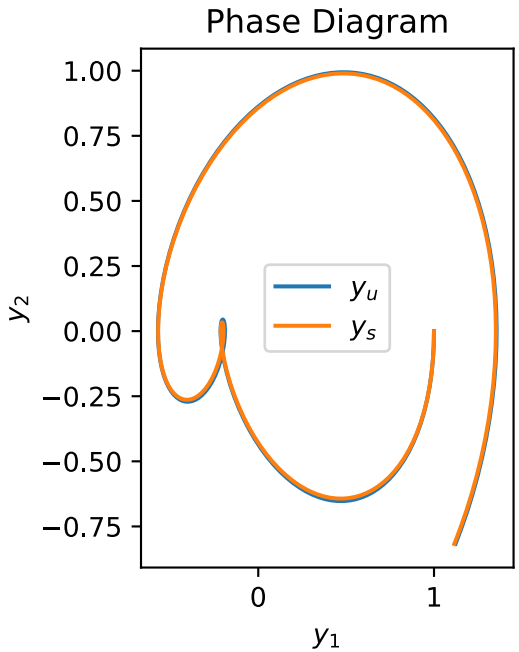


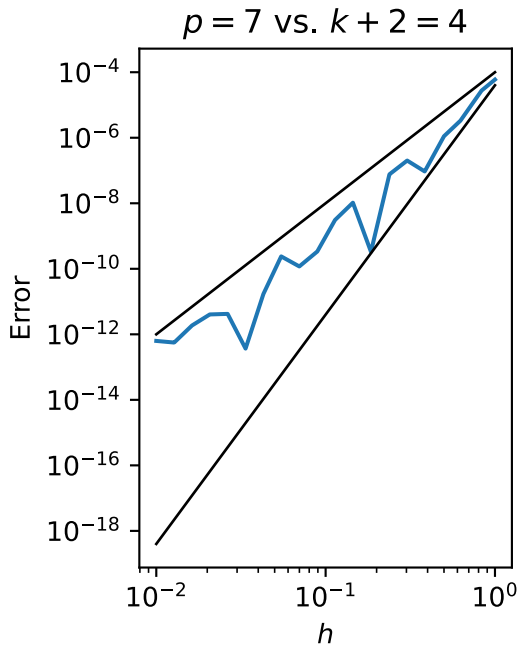


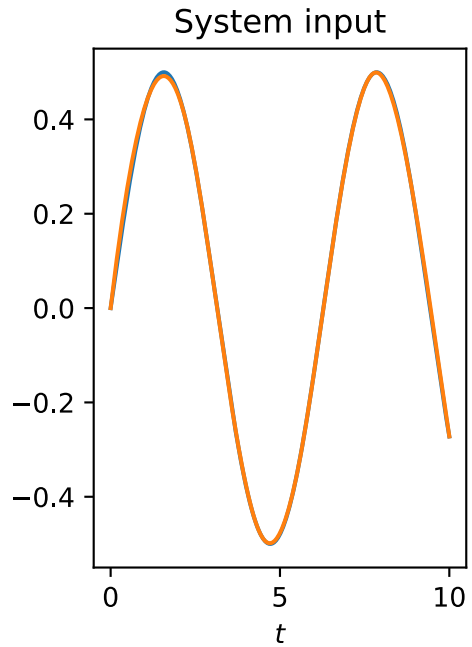
$$y_1(0) = 1,\ y_1'(0) = 0,\ y_1' = y_2, \alpha = 0.5, \omega = 0.3, t_0 = 0, t_e = 10.$$

Excited Oscillator, 7th order RK, cubic spline

$$y_1''(t) + y_1(t) = u(t),\ u(t) = \alpha \sin(\omega t), y_2 = y_1'.$$



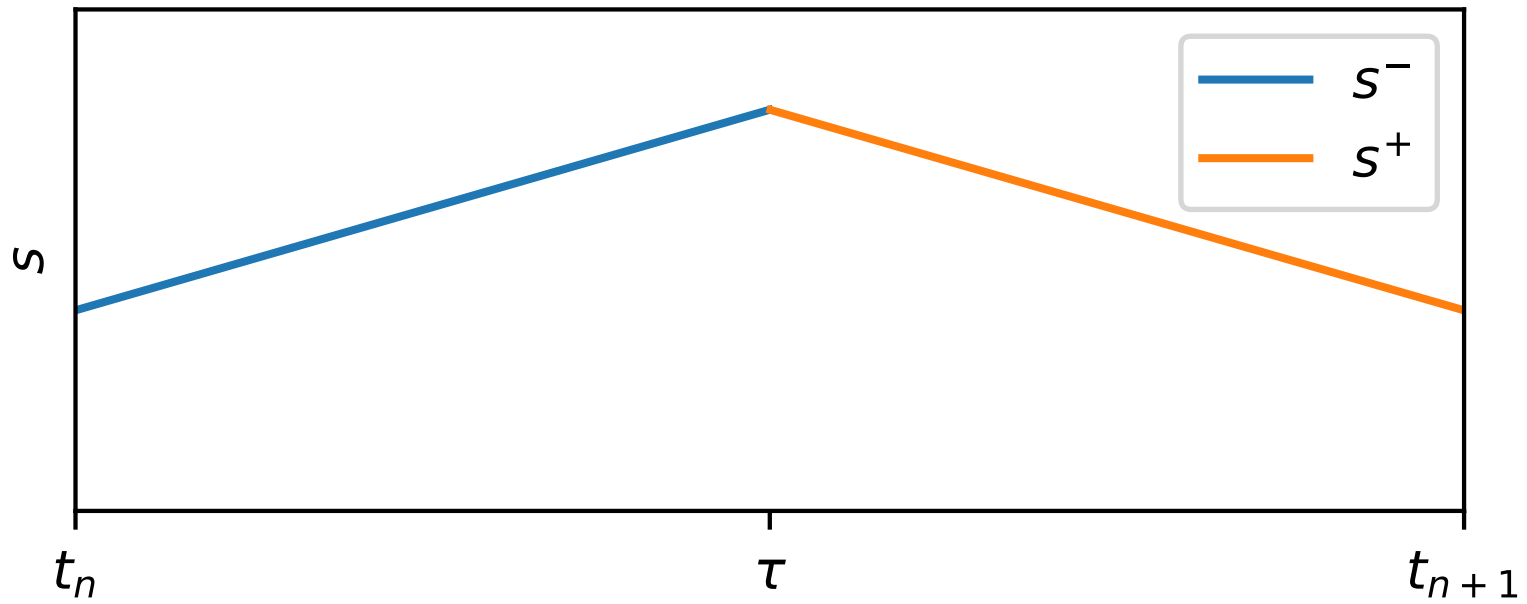




$$y_1(0) = 1,\ y_1'(0) = 0,\ y_1' = y_2, \alpha = 0.5, \omega = 1, t_0 = 0, t_e = 10.$$



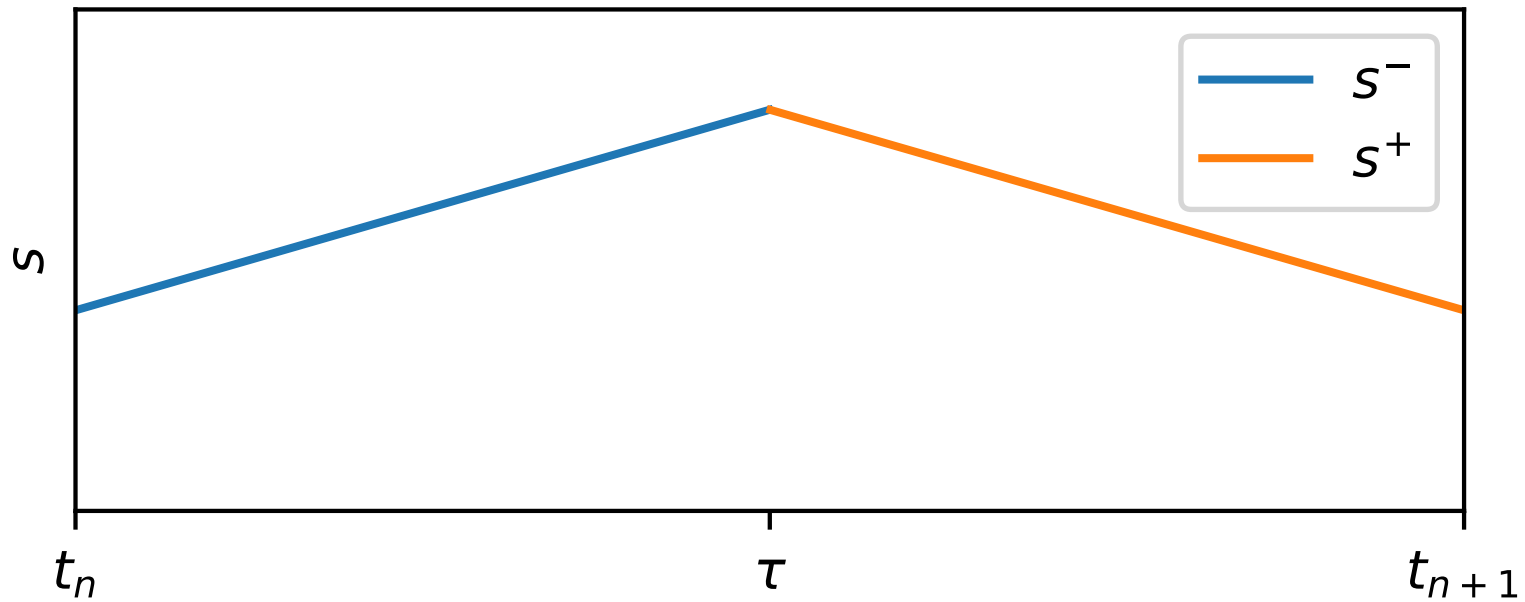
Proof sketch (1/3)



- Assume single discontinuity τ in step $t_n \to t_{n+1}$



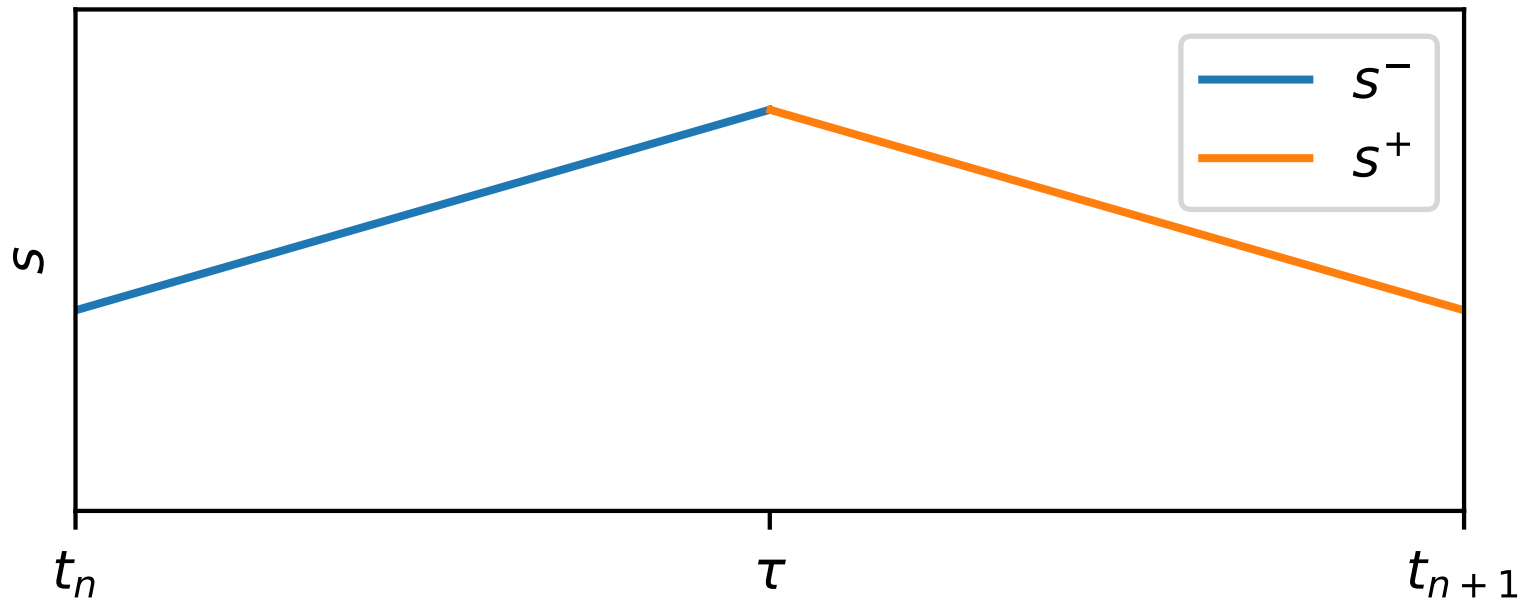
Proof sketch (1/3)



- Assume single discontinuity τ in step $t_n \to t_{n+1}$
- Left and right Taylor polynomials $s^-(t), s^+(t)$ around τ



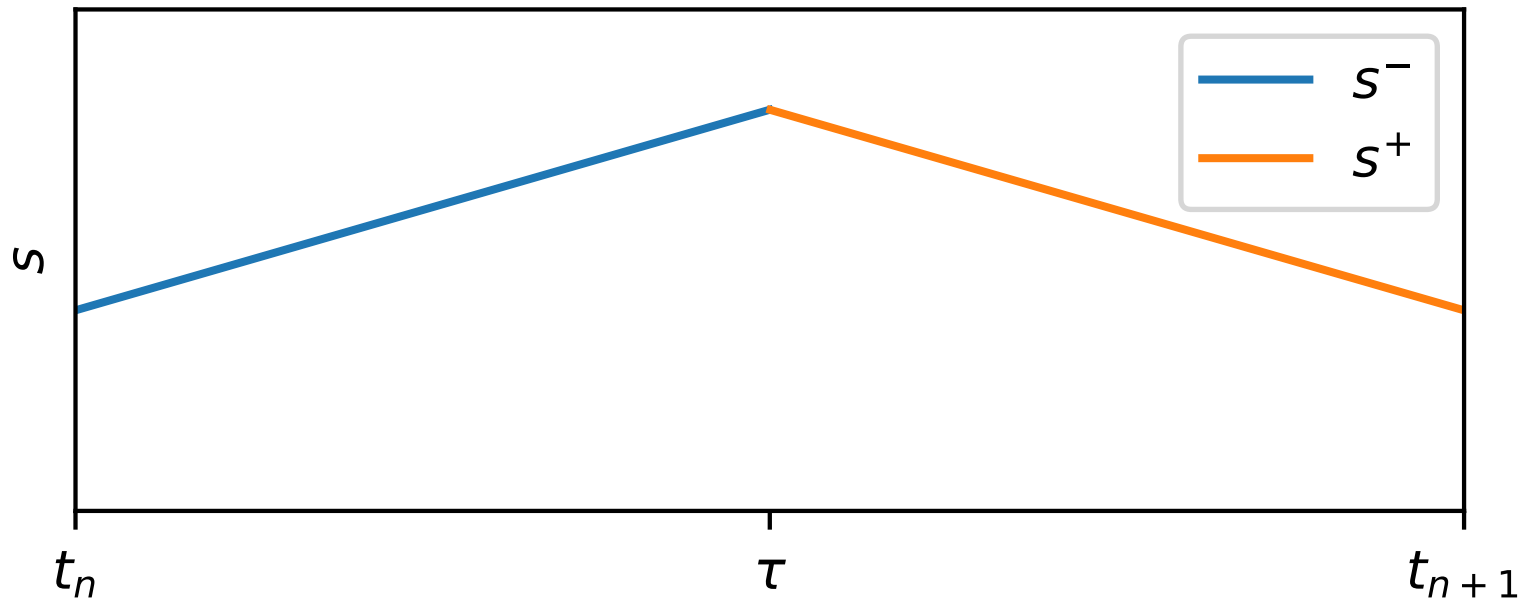
Proof sketch (1/3)



- Assume single discontinuity τ in step $t_n \to t_{n+1}$
- Left and right Taylor polynomials $s^-(t), s^+(t)$ around τ
- Continue each on $[t_n, t_{n+1}]$



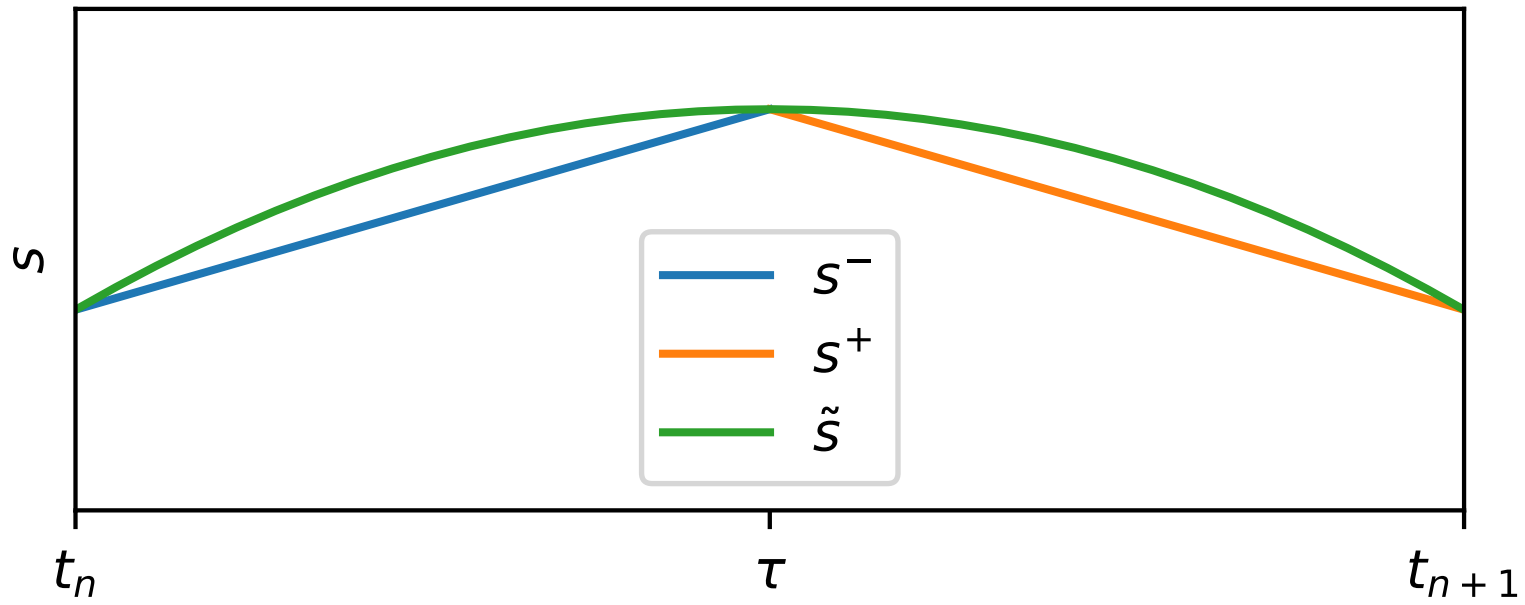
Proof sketch (1/3)



- Assume single discontinuity τ in step $t_n \to t_{n+1}$
- Left and right Taylor polynomials $s^-(t), s^+(t)$ around τ
- Continue each on $[t_n, t_{n+1}]$
- $\|s^+(t) - s^-(t)\| \leq 2Lh^{k+1}/((k+1)!)$

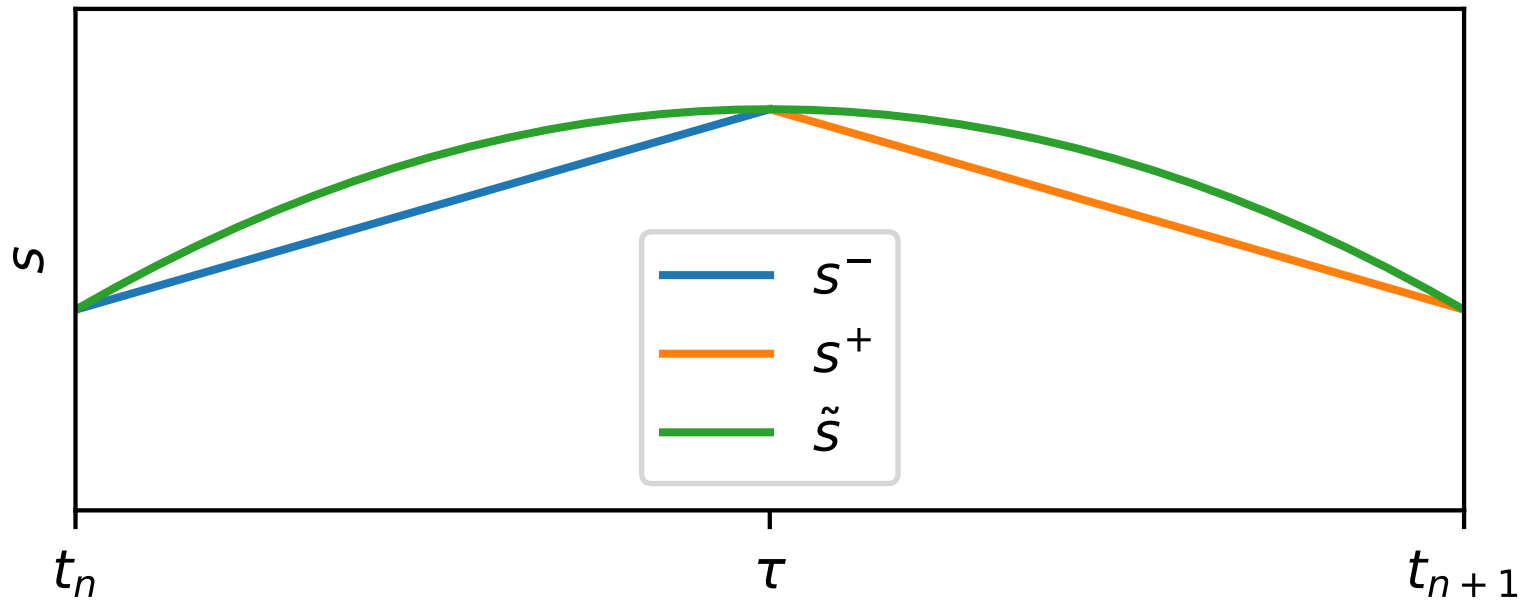


Proof sketch (2/3)



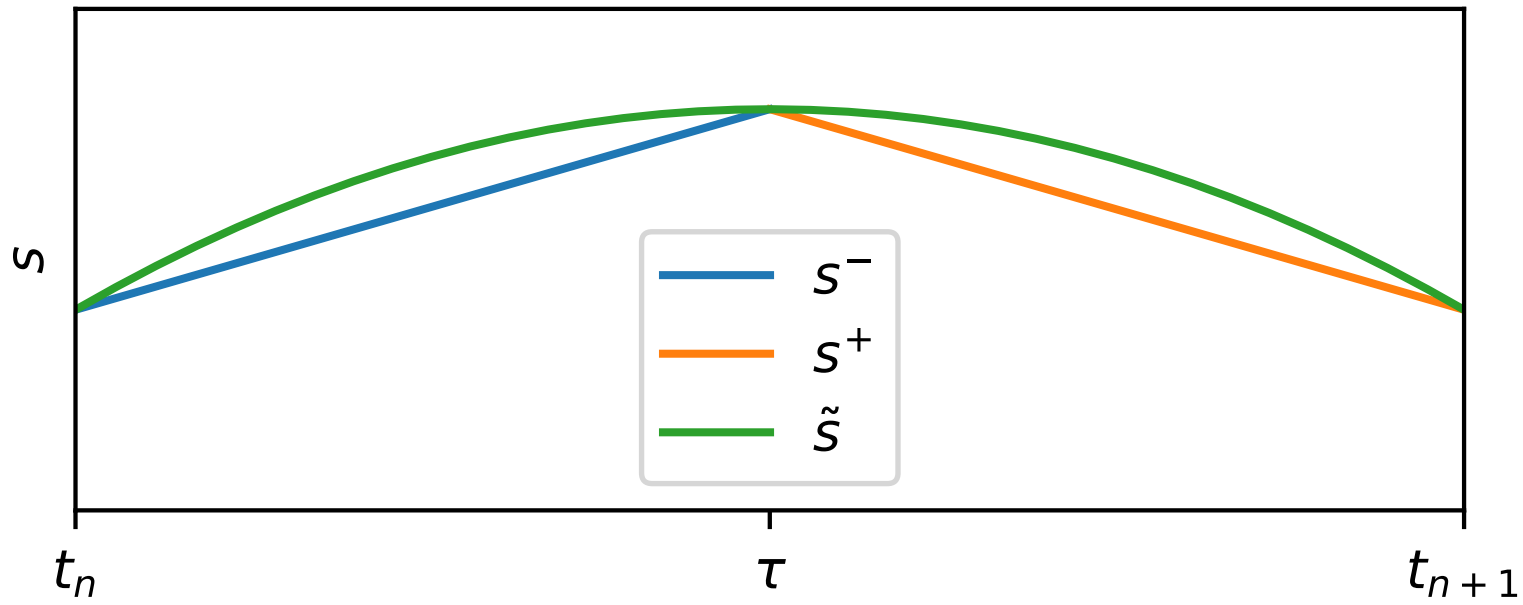
- Define smoothed system input
  $\tilde{s}(t) = (1-\lambda)s^-(t) + \lambda s^+(t), \quad \lambda = (t-t_n)/(t_{n+1}-t_n)$

Proof sketch (2/3)



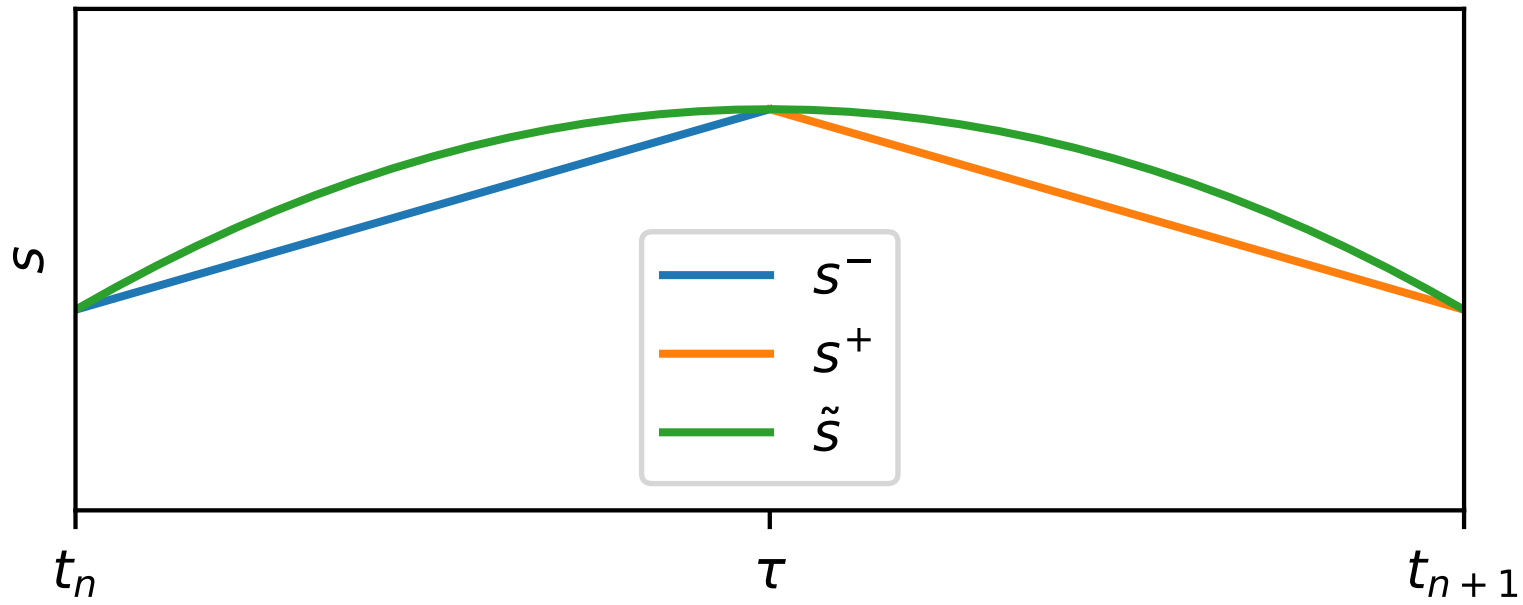
- Define smoothed system input
  $\tilde{s}(t) = (1-\lambda)s^-(t) + \lambda s^+(t), \quad \lambda = (t-t_n)/(t_{n+1}-t_n)$
- $\|\tilde{s}(t) - s(t)\| \leq 2Lh^{k+1}/((k+1)!)$

Proof sketch (2/3)



- Define smoothed system input
  $\tilde{s}(t) = (1-\lambda)s^-(t) + \lambda s^+(t), \quad \lambda = (t-t_n)/(t_{n+1}-t_n)$
- $\|\tilde{s}(t) - s(t)\| \leq 2Lh^{k+1}/((k+1)!)$
- Local, smooth shadow problem
  $\bar{y}'_{\tilde{s}}(t) = f(t, \bar{y}_{\tilde{s}}(t), \tilde{s}(t)), \bar{y}(t_n) = y(t_n)$

Proof sketch (2/3)



- Define smoothed system input
  $\tilde{s}(t) = (1-\lambda)s^-(t) + \lambda s^+(t), \quad \lambda = (t-t_n)/(t_{n+1}-t_n)$
- $\|\tilde{s}(t) - s(t)\| \leq 2Lh^{k+1}/((k+1)!)$
- Local, smooth shadow problem
  $\bar{y}'_{\tilde{s}}(t) = f(t, \bar{y}_{\tilde{s}}(t), \tilde{s}(t)), \bar{y}(t_n) = y(t_n)$
- $\|\bar{y}_{\tilde{s}}(t) - y_s(t)\| \leq C_0(e^{L_{f,s}h} - 1)h^{k+1} \leq Ch^{k+2}$
  (asymptotically)

Proof sketch (3/3)

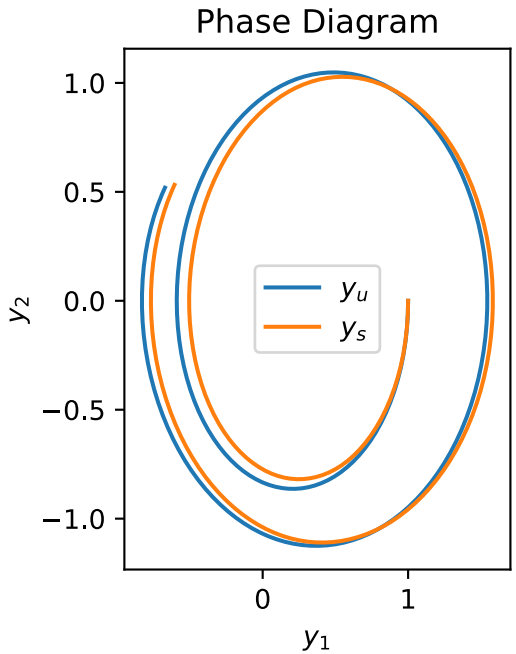
Result ($t_n \to t_{n+1}, \tau \in [t_n, t_{n+1}]$):

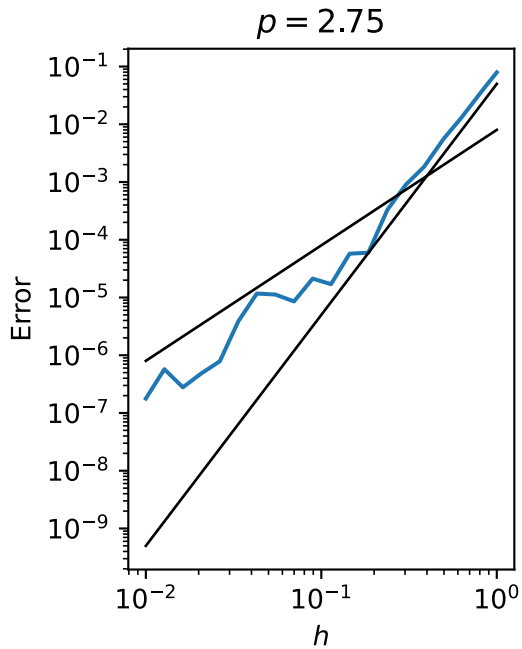
$$
\begin{aligned}
\|le_n\| &= \|y_s(t_{n+1}) - (y_s(t_n) + h\Phi(y_s(t_n); h, f, s))\| \\
&= \|y_s(t_{n+1}) \\
&\quad - (y_s(t_n) + h\Phi(y_s(t_n); h, f, \tilde{s})) \\
&\quad + (y_s(t_n) + h\Phi(y_s(t_n); h, f, \tilde{s})) \\
&\quad - \bar{y}_{\tilde{s}}(t_{n+1}) \\
&\quad + \bar{y}_{\tilde{s}}(t_{n+1}) \\
&\quad - (y_s(t_n) + h\Phi(y_s(t_n); h, f, s))\| \\
&\leq Ch^{k+2} + \mathcal{O}(h^{p+1}) + L_{\Phi,s}h^{k+2} \\
&= \mathcal{O}(h^{p+1}) + \mathcal{O}(h^{k+2}).
\end{aligned}
$$

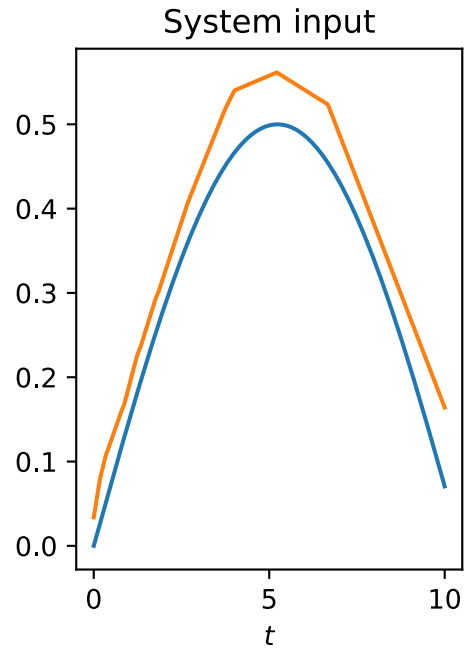


Excited Oscillator, 4th order RK, ReLU network s

$$y_1''(t) + y_1(t) = u(t),\ u(t) = \alpha \sin(\omega t).$$





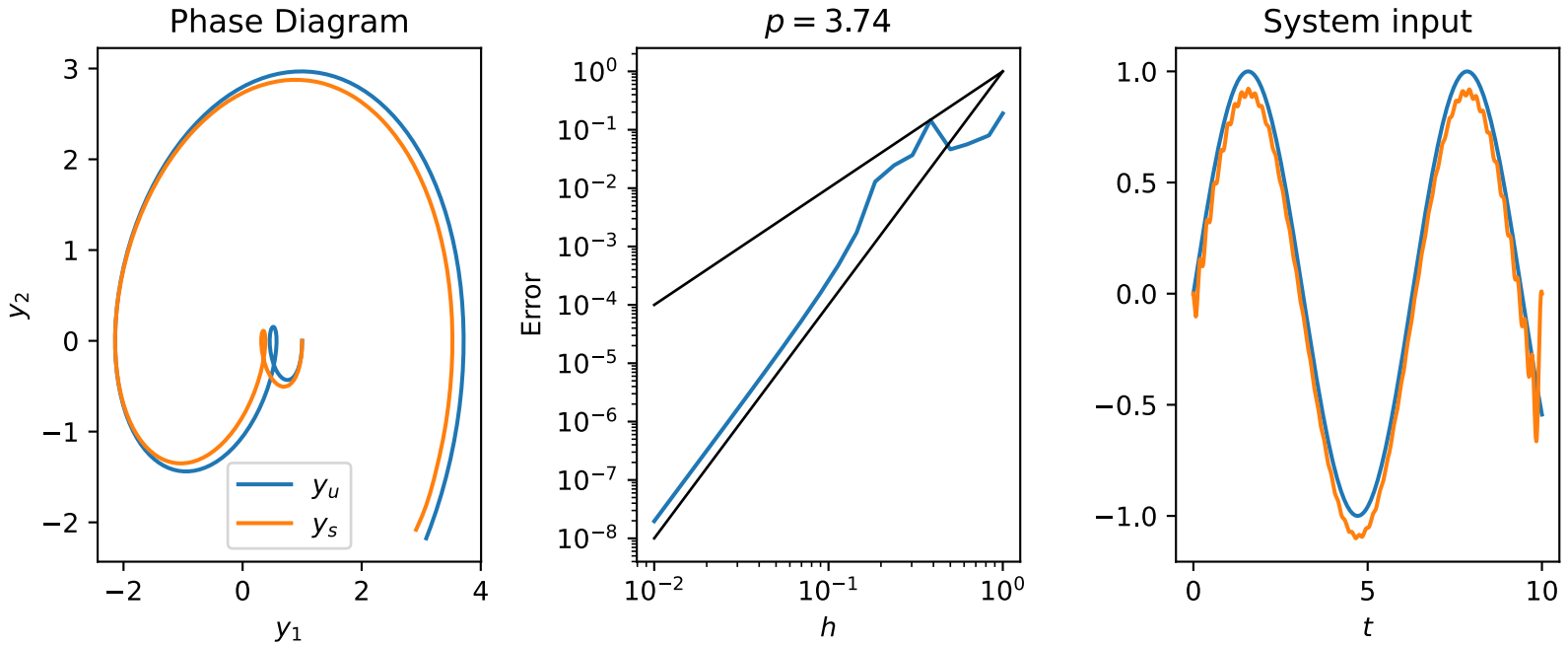


$$y_1(0) = 1,\ y_1'(0) = 0,\ y_1' = y_2, \alpha = 0.5, \omega = 0.3, t_0 = 0, t_e = 10.$$



Excited Oscillator, 4th order RK, Fourier s

$$y_1''(t) + y_1(t) = u(t),\ u(t) = \alpha \sin(\omega t).$$



$$y_1(0) = 1,\ y_1'(0) = 0,\ y_1' = y_2, \alpha = 1, \omega = 1, t_0 = 0, t_e = 10.$$

Generalization of previous theorem

Theorem

The numerical integration of

$$y'(t) = f(t, y(t), s(t)), \quad y(t_0) = y_0,$$

where s is reasonably non-smooth yields local error estimates

$$\|le_n\| = \mathcal{O}(h^{p+1}) \text{ and } \|le_n\| = \mathcal{O}(h^{p+1}) + \mathcal{O}(h^{k+2}).$$

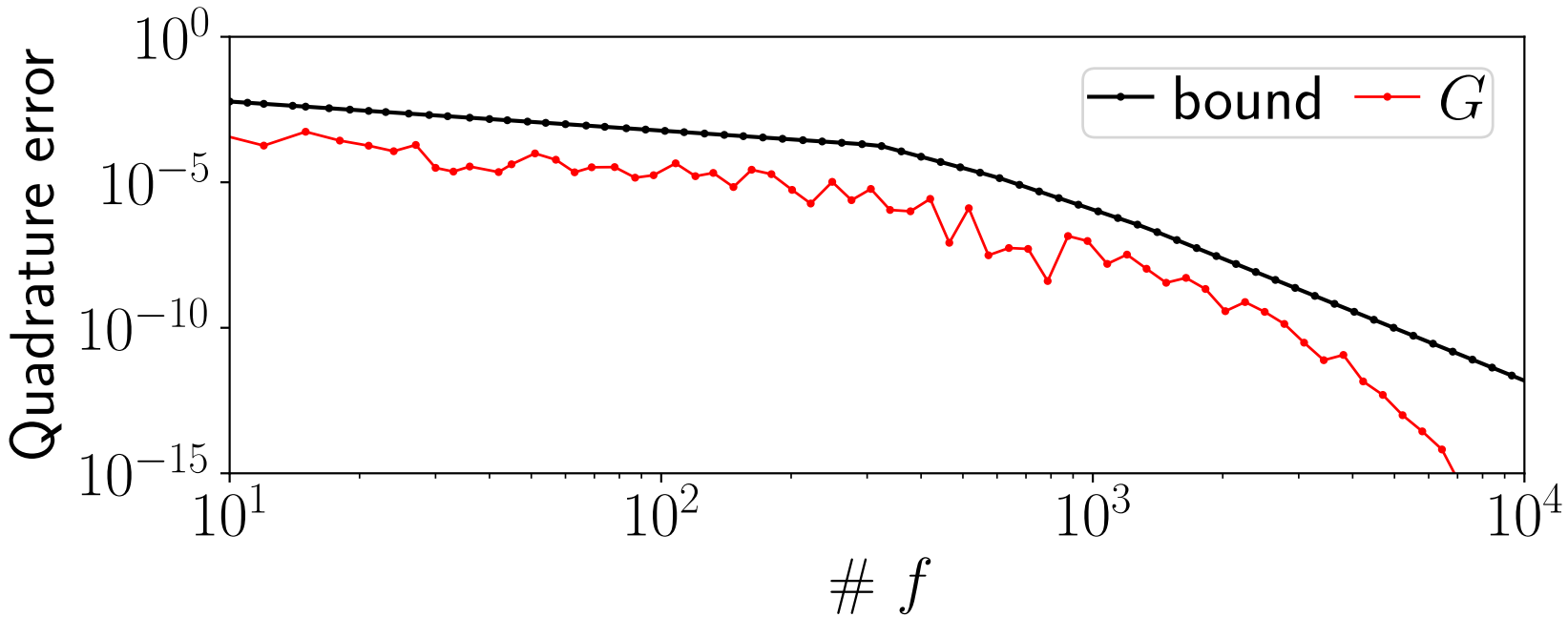
and as before

$$\|e_n\| = \mathcal{O}(h^p) + \mathcal{O}(h^{k+2}).$$

Refined error bound

A "NeuralODE" (NeuralQuadrature) for solving Euler's elastica using 3-point Gaussian quadrature.

$$\left| \int_a^b f(x)dx - \hat{\mathrm{I}}(f; a, b, h) \right| \leq \min_{r \leq k} \{ C_r \| f^{(r)} \|_\infty h^r \}$$



Celledoni, E.; Cokaj, E.; Leone, A.; Leyendecker, S.; Murari, D.; Owren, B.; Sato Martin de Almagro, R.; Stavole, M. (2024): Neural networks for the approximation of Euler's elastica". CMAME 435.117584.

State-dependent system input

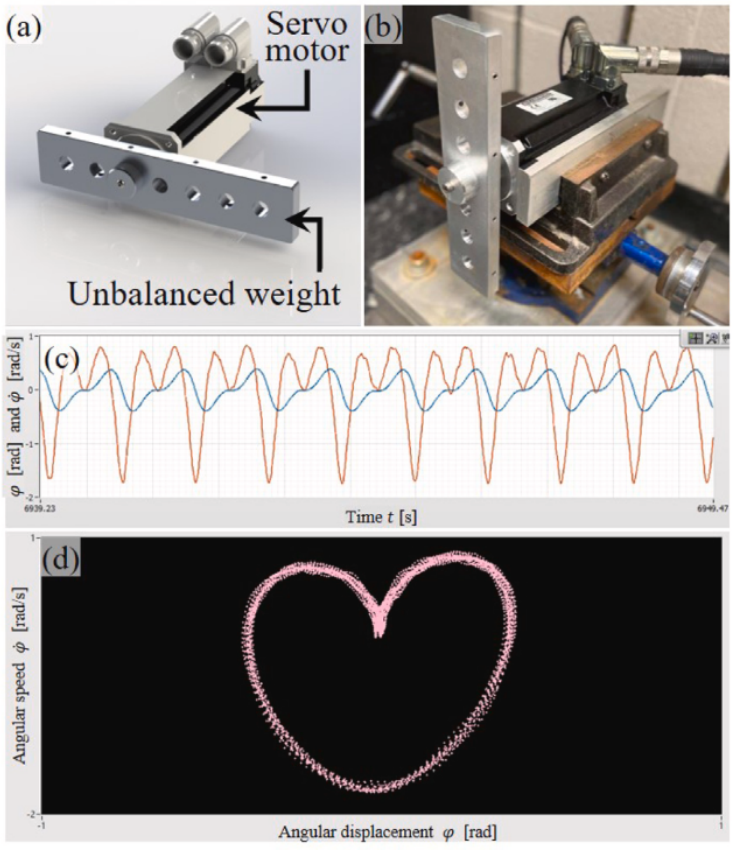
Rotating machinery, torsional dynamics:

- Lathe with uneven workpiece
- Drilling anisotropic rocks (oil and gas exploration)
- Rub-impact rotors

Liu, X; Hou, X:, Meng, G. (2026): Dynamics of a nonlinear oscillator with space-time phase-modulated excitation. International Journal of Non-Linear Mechanics, 187.

State-dependent system input

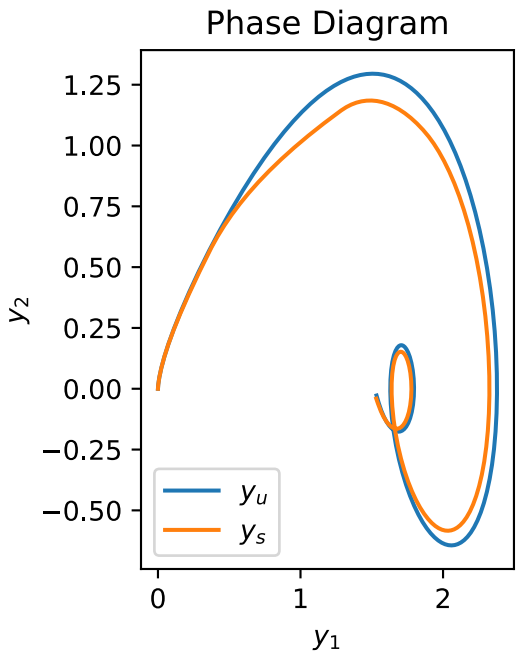
$$m y_1''(t) + \alpha y_1(t) + \gamma y_1'(t) = u(t, y_1(t))$$

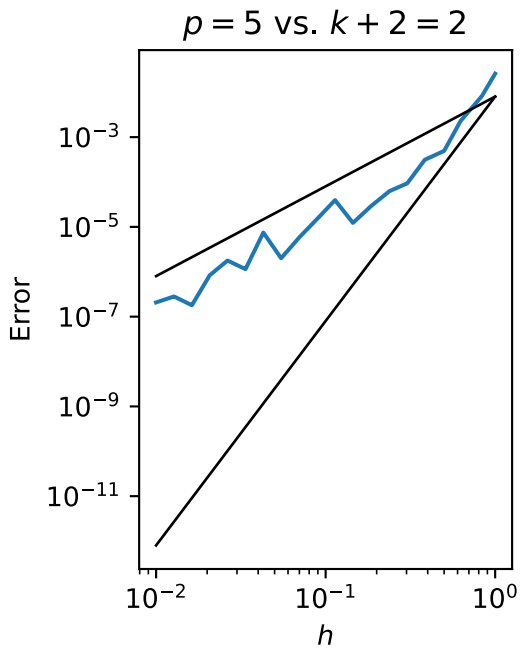


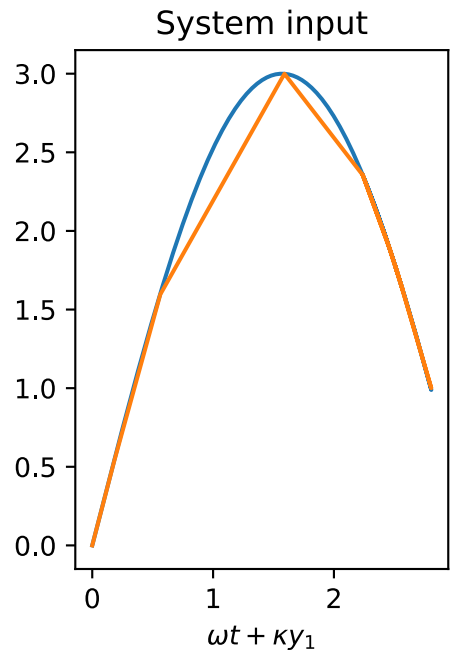
Liu, X; Hou, X:, Meng, G. (2026): Dynamics of a nonlinear oscillator with space-time phase-modulated excitation. International Journal of Non-Linear Mechanics, 187.

State-dependent system input, linear spline

$$y_1''(t) + \alpha y_1(t) + \gamma y_1'(t) = \beta \sin(\omega t + \kappa y_1(t))$$



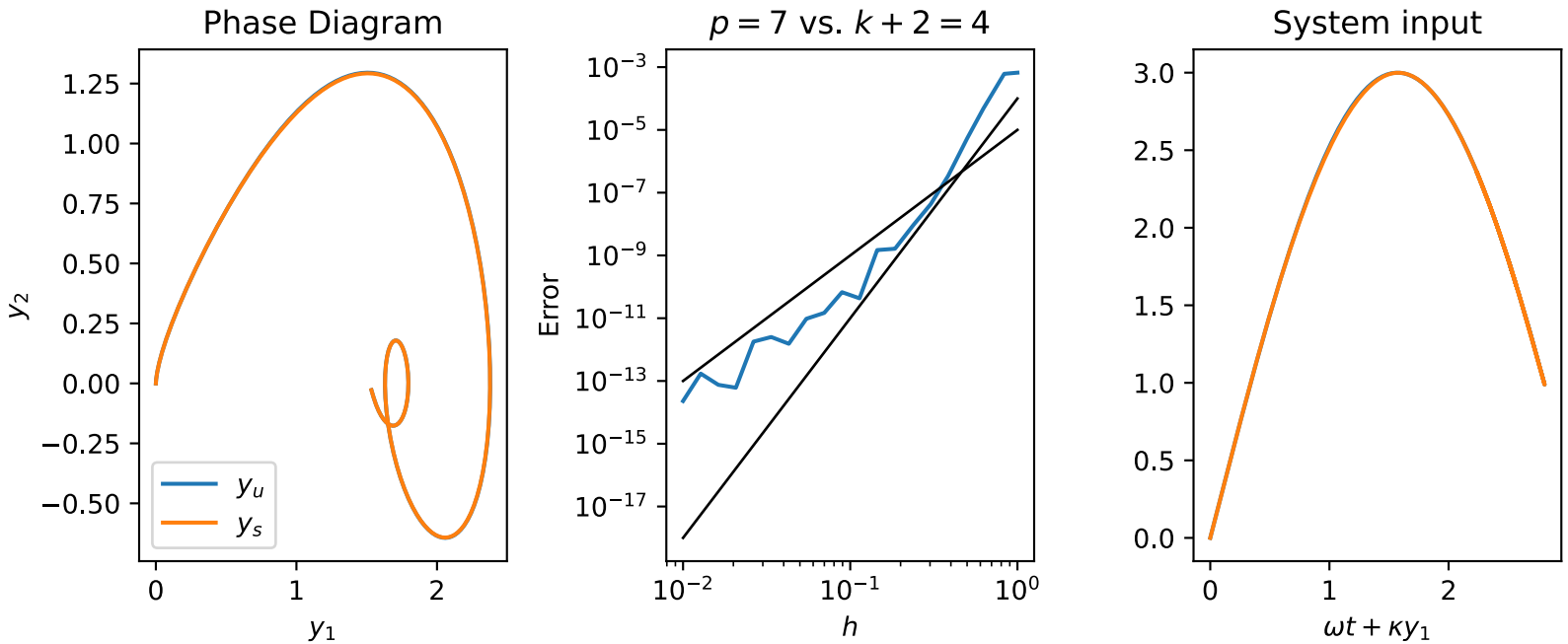




$$\alpha = 1.104, \beta = 3, \gamma = 0.958, \omega = 0.1, \kappa = 1, t_0 = 0, t_e = 10.$$

State-dependent system input, cubic spline

$$y_1''(t) + \alpha y_1(t) + \gamma y_1'(t) = \sin(\omega t + \kappa y_1(t))$$



$$\alpha = 1.104, \beta = 3, \gamma = 0.958, \omega = 0.1, \kappa = 1, t_0 = 0, t_e = 10.$$



Natural hypothesis

Hypothesis

The numerical integration of

$$y'(t) = f(t, y(t), s(y(t))), \quad y(t_0) = y_0,$$

where s is reasonably non-smooth yields local error estimates

$$\|le_n\| = \mathcal{O}(h^{p+1}) \text{ and } \|le_n\| = \mathcal{O}(h^{p+1}) + \mathcal{O}(h^{k+2}).$$

and as before the global error estimate

$$\|e_n\| = \mathcal{O}(h^p) + \mathcal{O}(h^{k+2}).$$

Proof Sketch

Different to before:

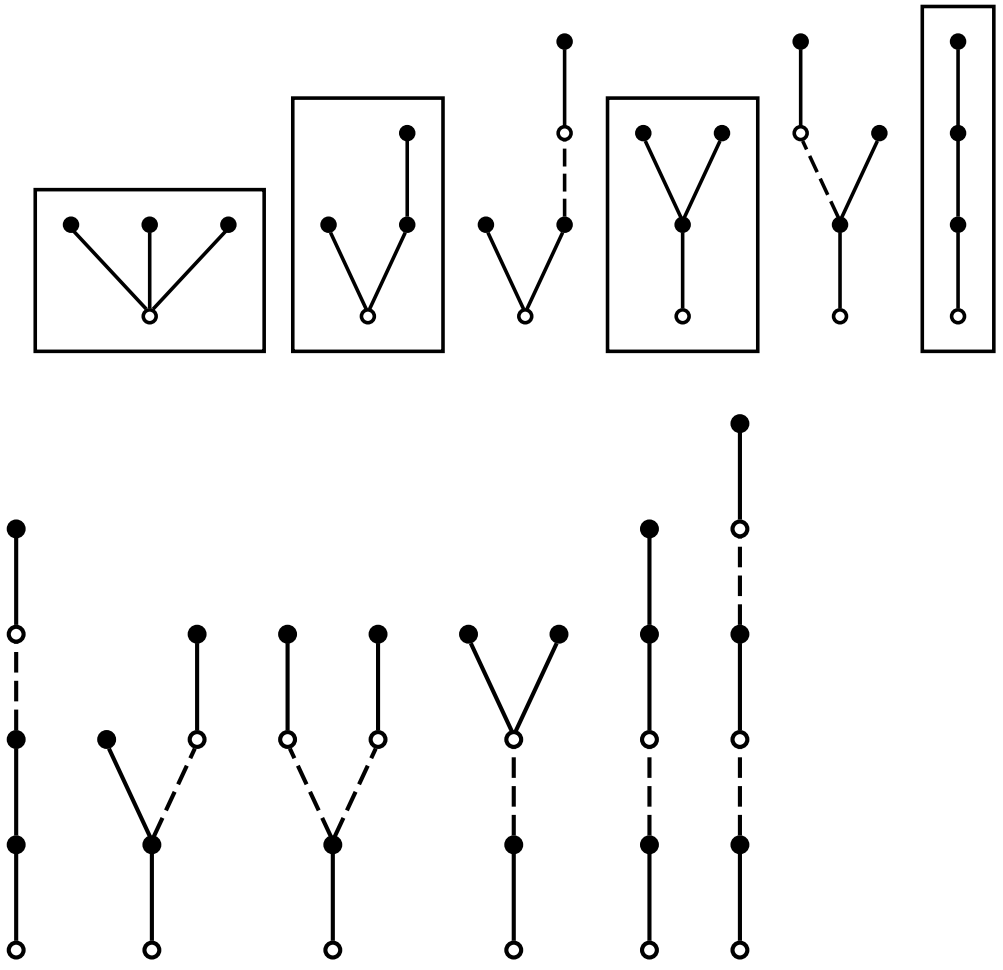
- Assume $[t_n, t_{n+1}]$ contains at most one τ_i, such that $s(y(\tau_i))$ discontinuous (quite restricting)
- Left and right Taylor polynomials

$$s^-(y(t)) = s(y(\tau_i)) + (t - \tau_i)s_y(y(\tau_i))f(y(\tau_i), s(y(\tau_i))) + \ldots,$$
$$s^+(y(t)) = s(y(\tau_i)) + (t - \tau_i)s_y(y(\tau_i))f(y(\tau_i), s(y(\tau_i))) + \ldots,$$

$\rightarrow$ bound on $\frac{d^{k+1}}{dt^{k+1}} s(y(t))$
$\rightarrow$ generalized elementary differentials



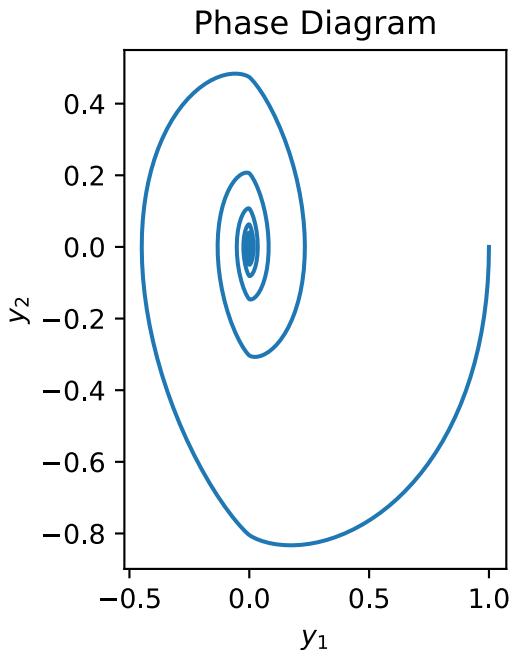
Example: Corresponding Trees of order 4

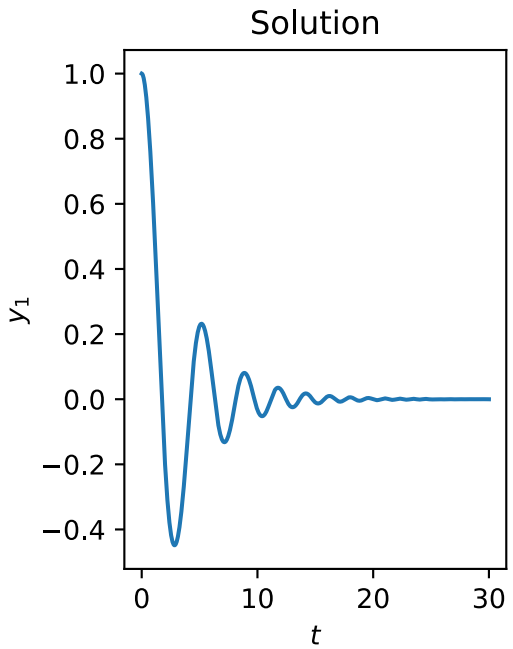


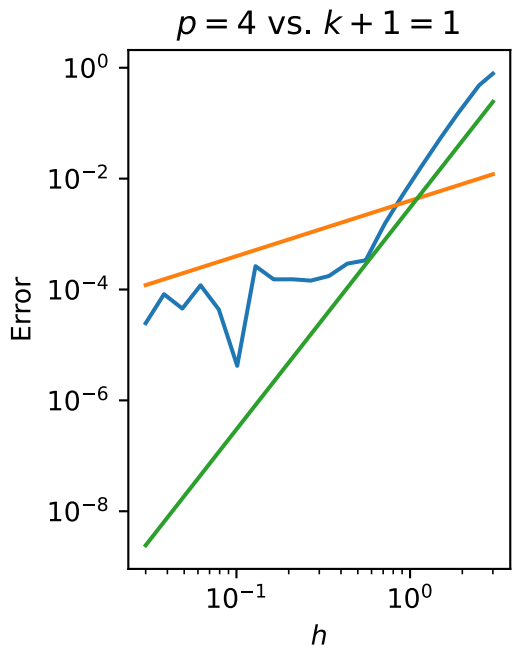


Unavoidable multiplicity of discontinuities in step $t_n \to t_{n+1}$

Finitely many $\to$ Similiar result, growing error constants
Infinitely many $\to$ Equilibria or infinite oscillations in finite time $\to$ Loss of one order?







$$y_1'' + \operatorname{sign}(y_1)\sqrt{|y_1|} + 0.5y_2 = 0,\ y_1(0) = 1,\ y_2'(0) = 0, y_2 = y_1'.$$

Conclusion

- Error bounds for spline $s = s(t)$ directly transferable to other reasonably non-smooth $s = s(t)$
- Refined error bound useful in oscillatory context
- Also transferable to state-dependent $s = s(y(t))$, but with more restrictive assumptions
- Experimentally, without assumptions, loss of one order seems to be the case, but more work needed

